



Unit No 2

Understanding the Design Process

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Unit -2 at a Glance



Unit-II: Understanding the Design Process

- Interaction design basics.
- HCI in the software process, Design rules, Implementation support, Evaluation techniques, Universal design, User support, Individual differences, designing interfaces for all, User research and techniques,
- Understanding Personae, Good and poor design, Ergonomics.
- **Exercise - Creating personae for different applications in everyday use.**



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Interaction design basics



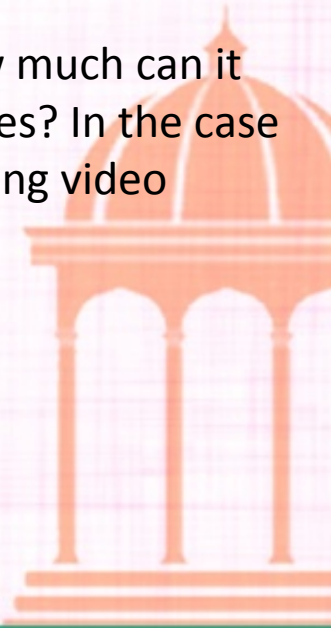
WHAT IS DESIGN?

So what is design? A simple definition is: **achieving goals within constraints**

This does not capture everything about design, but helps to focus us on certain things:

Goals: What is the purpose of the design we are intending to produce? Who is it for? Why do they want it? For example, if we are designing a wireless personal movie player, we may think about young affluent users wanting to watch the latest movies whilst on the move and download free copies, and perhaps wanting to share the experience with a few friends.

Constraints: What materials must we use? What standards must we adopt? How much can it cost? How much time do we have to develop it? Are there health and safety issues? In the case of the personal movie player: **does it have to withstand rain?** Must we use existing video standards to download movies? Do we need to build in copyright protection?



WHAT IS DESIGN?

Of course, we cannot always achieve all our goals within the constraints. So perhaps one of the most important things about design is:

Trade-off : Choosing which goals or constraints can be relaxed so that others can be met. For example, we might find that an eye-mounted video display, a bit like those used in virtual reality, would give the most stable image whilst walking along. However, this would not allow you to show friends, and might be dangerous if you were watching a gripping part of the movie as you crossed the road.

Often the most exciting moments in design are when you get a radically different idea that allows you to satisfy several apparently contradictory constraints

Overview



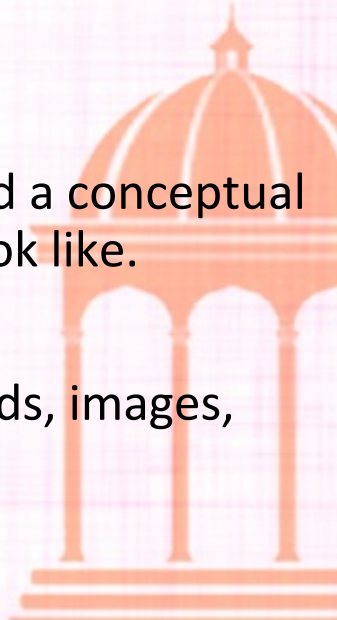
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- What is involved in Interaction Design?
 - Importance of involving users
 - Degrees of user involvement
 - What is a user-centered approach?
 - Four basic activities
- Some practical issues
 - Who are the users?
 - What are 'needs'?
 - Where do alternatives come from?
 - How do you choose among alternatives?
- A simple lifecycle model for Interaction Design
- Lifecycle models from software engineering
- Lifecycle models from HCI



Four Basic Activities of Interaction Design

- 1. Identifying needs and establishing requirements
 - In order to design something to support people, we must know who our target users are and what kind of support an interactive product could usefully provide.
- 2. Developing alternative designs
 - This is the core activity of designing: actually suggesting ideas for meeting the requirements.
 - Conceptual Design
 - Involves producing the conceptual model for the product, and a conceptual model describes what the product should do, behave, and look like.
 - Physical Design
 - Considers the detail of the product including the colors, sounds, images, menu design, and icon design.





Four Basic Activities of Interaction Design

- 3. Building interactive versions of the designs
 - The most sensible way for users to evaluate designs is to interact with them.
 - This does not mean that a software version is required, but rather, a paper-based prototype is quick and cheap to build.
- 4. Evaluating designs
 - Evaluation is the process of determining the usability and acceptability of the design.
 - Evaluation is measured in terms of a variety of criteria including:
 - numbers of errors users make using it
 - how appealing it is
 - how well it matches the requirements



Trans design process: what is a design?



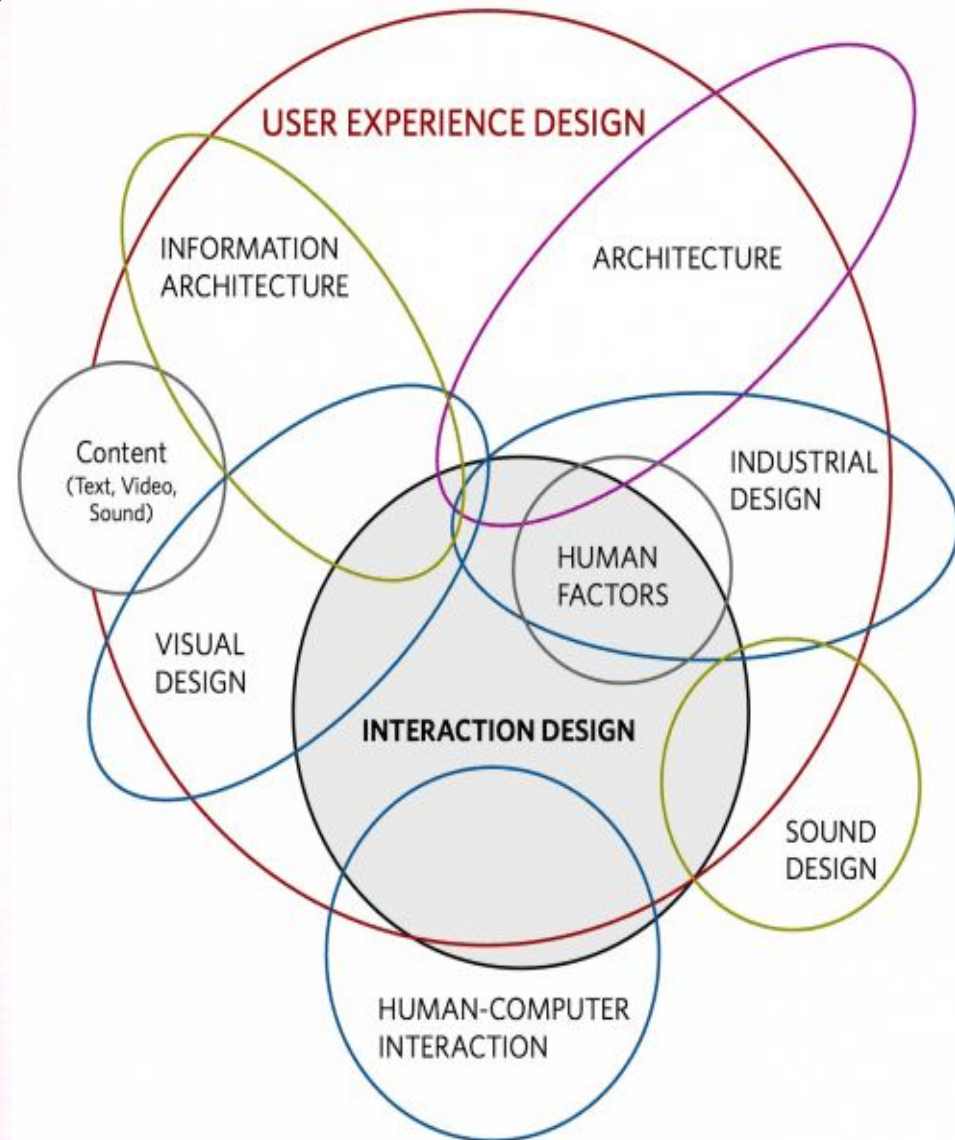
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What is interaction design?

Interaction design, often abbreviated as IxD, is the practice of designing interactive digital products and services.

Interaction Design is the creation of a dialogue between a person and a product, system, or service. This dialogue is both physical and emotional in nature and is manifested in the interplay between form, function, and technology as experienced over time."

Interaction designers focus on the way users interact with products and they use principles of good communication to create desired user experiences.



Trans design process: what is a design?

User Centered Design (UCD)

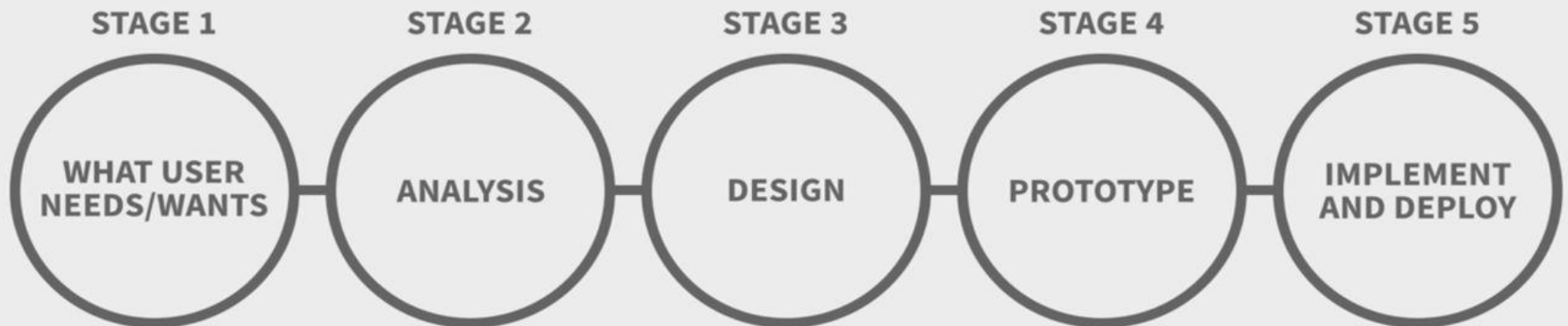
The process of collecting feedback from users to improve the design is known as *user centered design* or UCD.

UCD Drawbacks

- Passive user involvement.
- User's perception about the new interface may be inappropriate.
- Designers may ask incorrect questions to users.



GENERAL FLOW OF INTERACTION DESIGN PROCESS



Three Key Characteristics of the Interaction Design Process

- 1. A User Focus

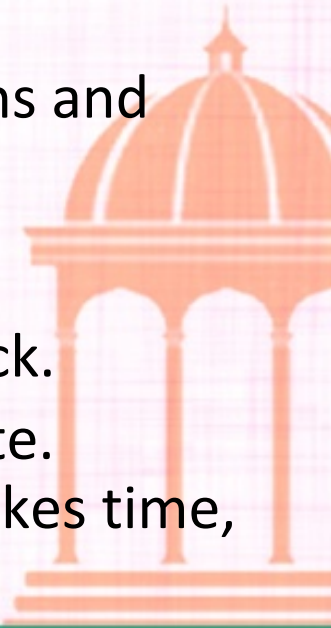
- A process cannot guarantee that a development will involve users, it can encourage focus on such issues and provide opportunities for evaluation and user feedback.

- 2. Specific Usability Criteria

- Specific usability and user experience goals should be identified, clearly documented, and agreed upon and the beginning of the project.
- They help designers choose between alternative designs and check on progress.

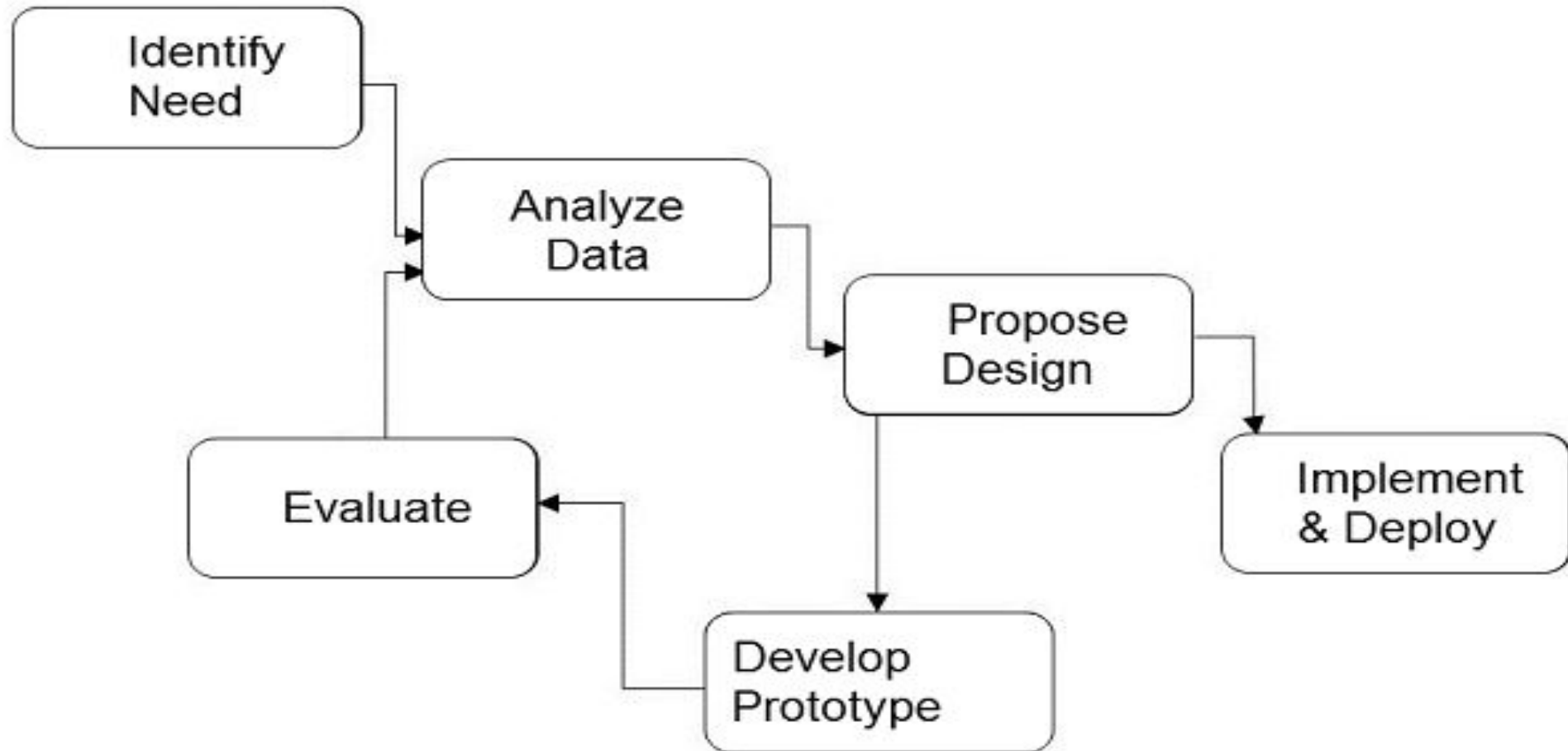
- 3. Iteration

- Iteration allows designs to be refined based on feedback.
- Iteration is important useful if you are trying to innovate. Innovation rarely emerges whole and ready to go. It takes time, evolution, trial and error, and patience.



Trans design process: what is a design?

Interactive System Design Life Cycle (ISLC)





HCI in the software process

- Software engineering and the design process for interactive systems
- Usability engineering
- Iterative design and prototyping
- Design rationale



The software lifecycle

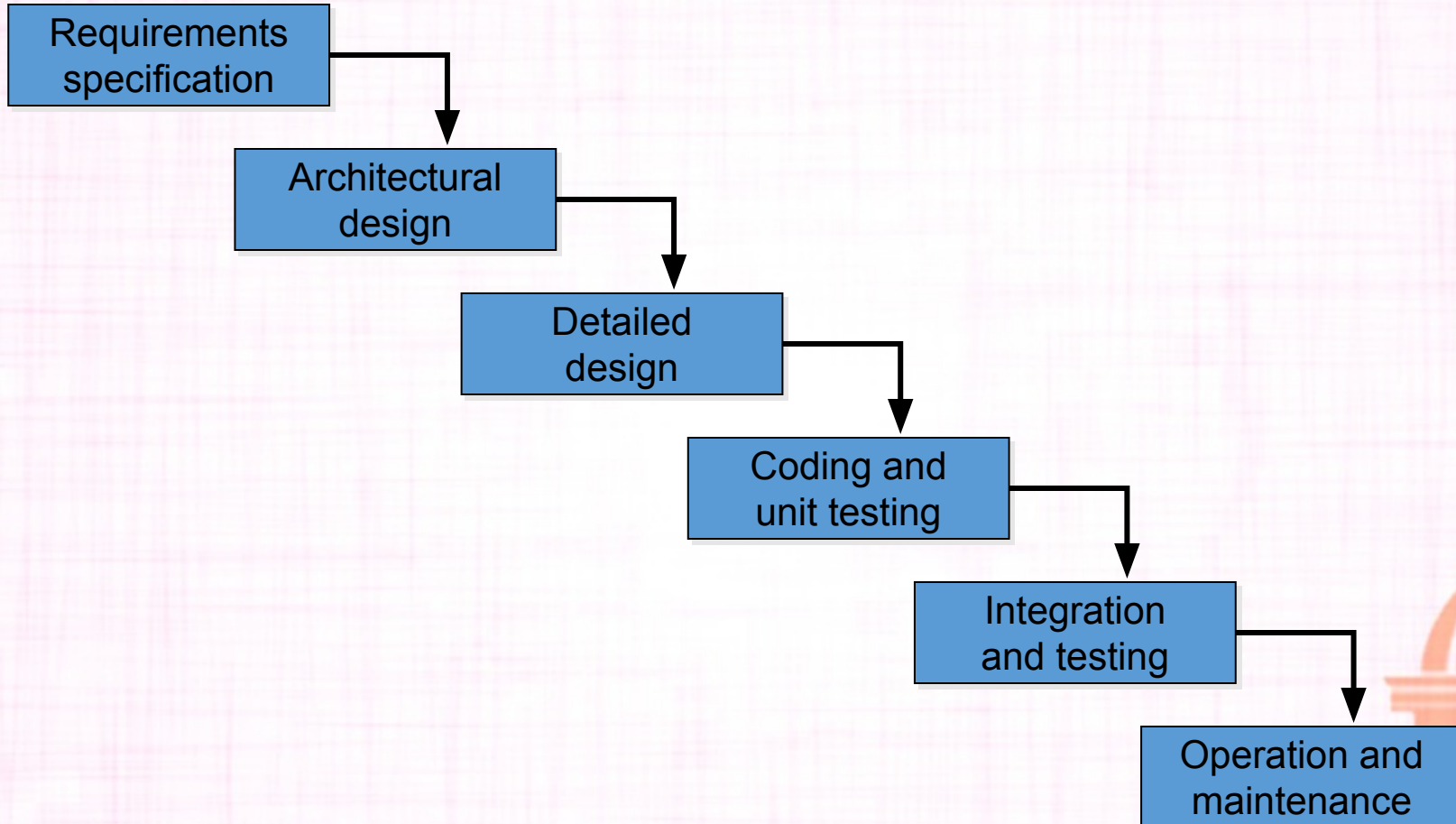
- Software engineering is the discipline for understanding the software design process, or life cycle
- Designing for usability occurs at all stages of the life cycle, not as a single isolated activity
- There are many models of the software life cycle we will look at the 2 main ones.
 - Waterfall, which assumes all the requirements can be predefined
 - Prototyping, which assumes the system must be build for the requirements to emerge.
- In reality most software development is somewhere between these two extremes



The waterfall model



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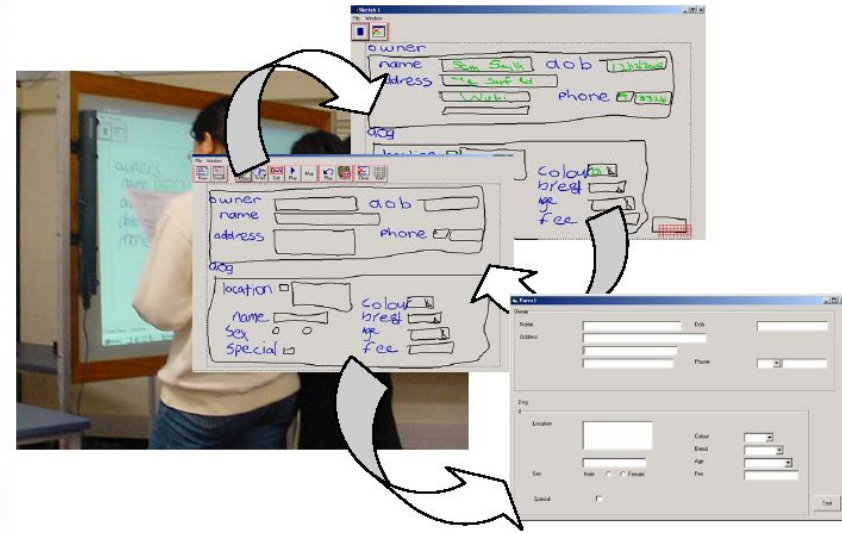
Activities in the life cycle

Requirements specification

designer and customer try capture what the system is expected to provide can be expressed in natural language or more precise languages, such as a task analysis would provide

Architectural design

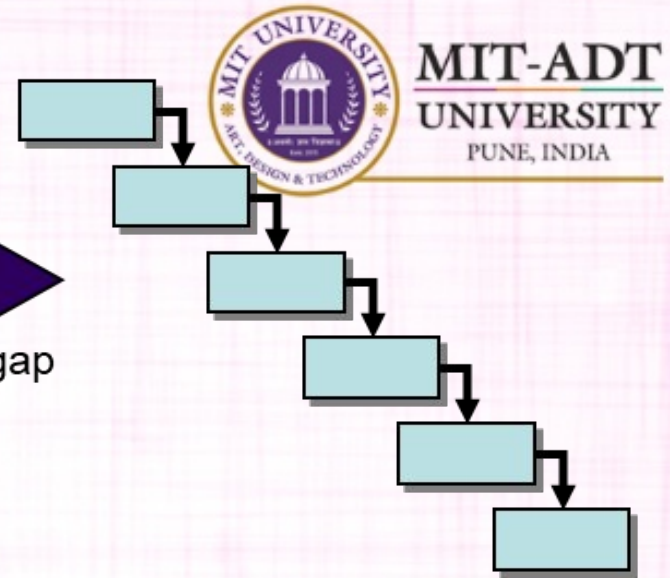
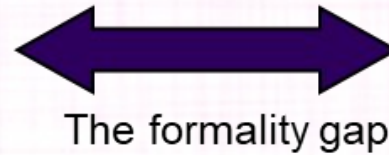
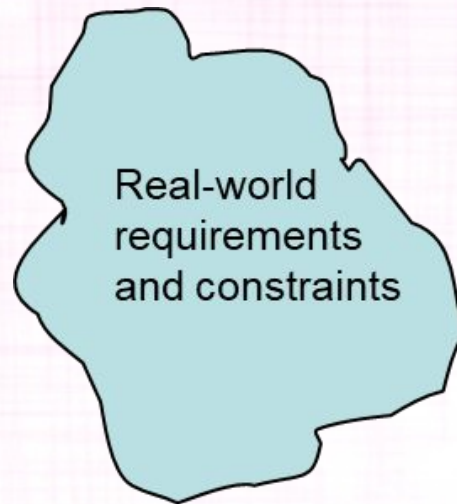
high-level description of how the system will provide the services required factor system into major components of the system and how they are interrelated needs to satisfy both functional and nonfunctional requirements



Detailed design

refinement of architectural components and interrelations to identify modules to be implemented separately the refinement is governed by the nonfunctional requirements

Verification and validation



Verification

designing the product right

Validation

designing the right product

The formality gap

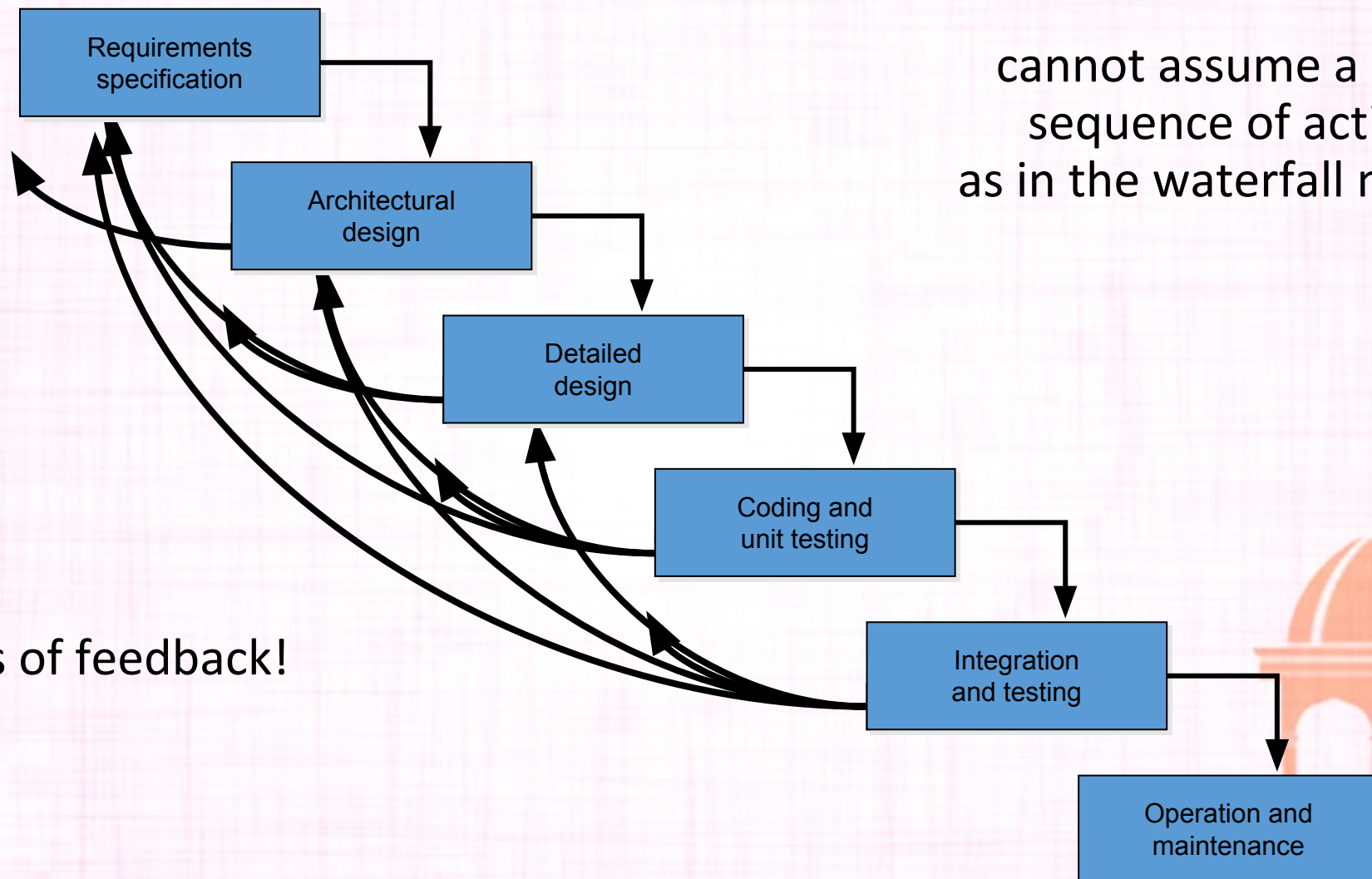
validation will always rely to some extent on subjective means of proof

Management and contractual issues

design in commercial and legal contexts



The life cycle for interactive systems



cannot assume a linear
sequence of activities
as in the waterfall model

lots of feedback!

Usability engineering

The ultimate test of usability based on measurement of user experience

Usability engineering demands that specific usability measures be made explicit as requirements

Usability specification

- usability attribute/principle

- measuring concept

- measuring method

- now level/ worst case/ planned level/ best case

Problems

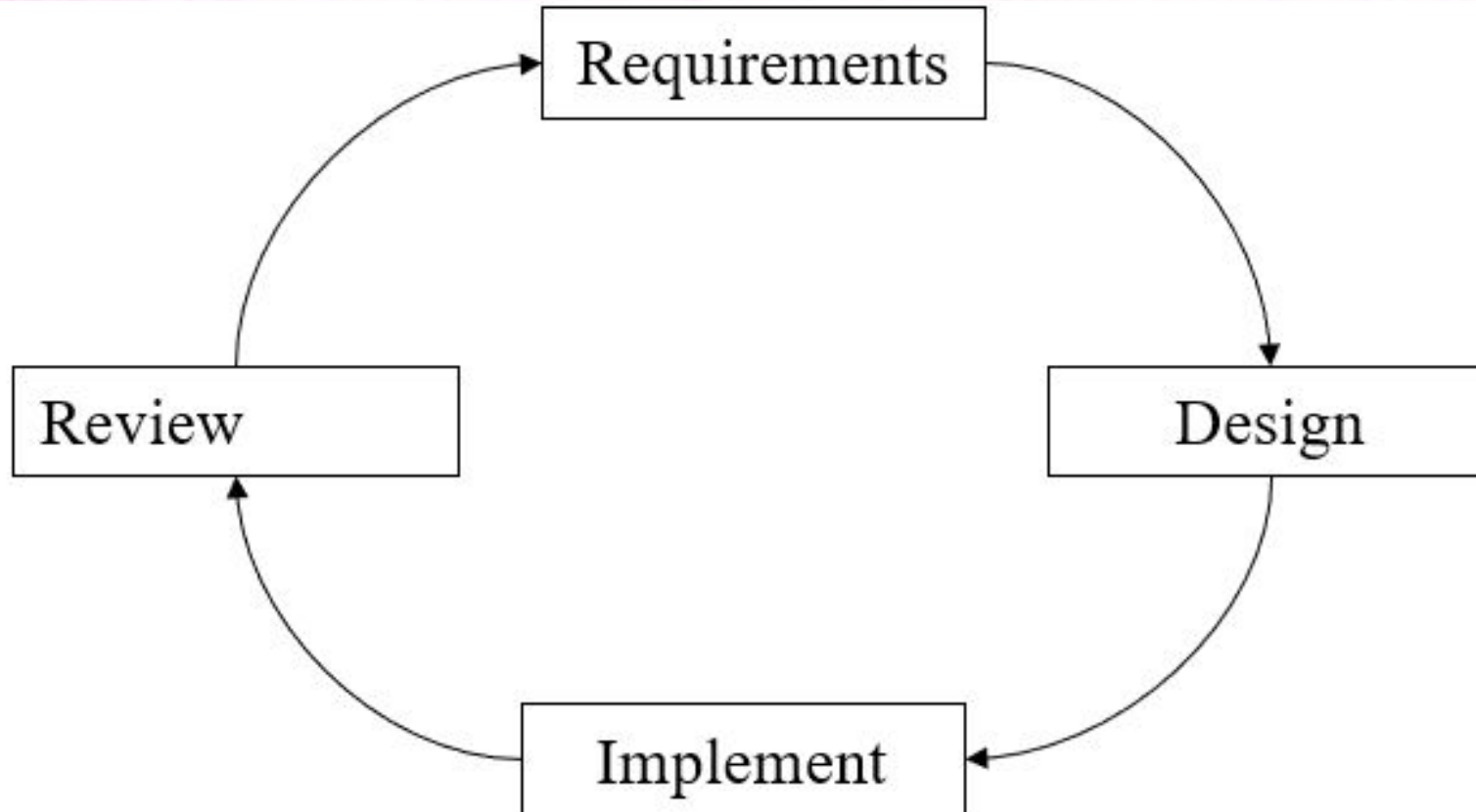
- usability specification requires level of detail that may not be possible early in design
- satisfying a usability specification does not necessarily satisfy usability



Iterative design and prototyping



- Iterative design overcomes inherent problems of incomplete requirements



Design rationale

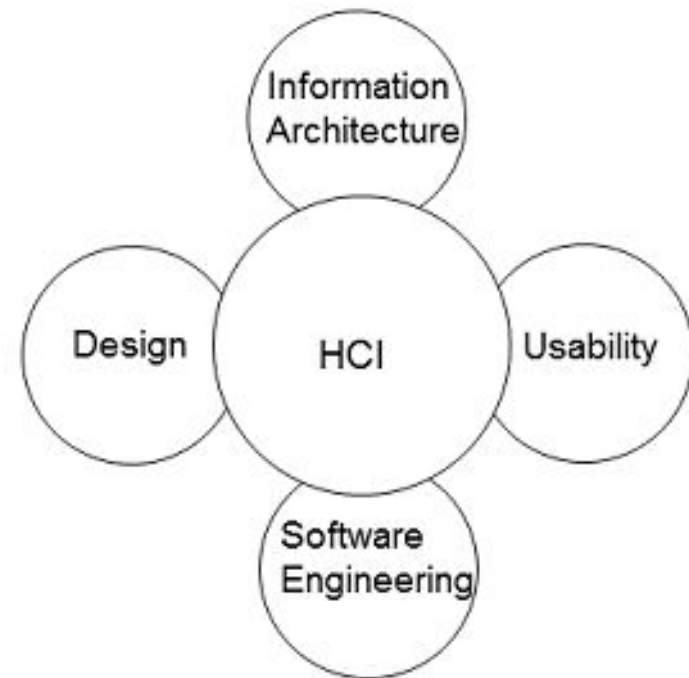
Design rationale is information that explains why a computer system is the way it is.

Benefits of design rationale

- communication throughout life cycle
- reuse of design knowledge across products
- enforces design discipline
- presents arguments for design trade-offs
- organizes potentially large design space
- capturing contextual information



Design Rules



HCI Design Rules and Best Practices



There are many **best practices and rules** that have been established by HCI researchers and designers. These rules include abstract design principles, to common-place standards, and design guidelines. Let's examine the most important design rules for HCI.





1. Design for learnability and familiarity

Learnability and familiarity principles determine how easily a novice user can learn to interact with a system. To achieve this, a system should be **predictable**. Any user should be able to predict the results of future actions based on the knowledge of their interaction history.

Similarly, a user should be able to map experiences with other computing systems to a new system. Familiarity determines whether a user can initiate interaction. Metaphors (usually visual) are a common way that systems achieve this principle.

For example, a trash bin icon should represent the action to delete items, as it is familiar and predictable.

2. Make elements legible and accessible



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All characters and objects must be perceptible in order to be used effectively. Multiple modes of “legibility” should be used to represent information (visual, verbal, and haptic). Similarly, a system must be equally useful to users with diverse abilities. For example, users with visual impairment should be provided with equivalent support systems.

3. Tolerance for error

Users will make errors, but they should not change the system state or impact an interaction disastrously. Warnings should be provided to prevent potential mistakes, and a system should not have a critical situation that can be reached easily by the user. A system must also offer assistance when potential errors arise.

4. Flexibility

Any design should provide multiple ways to perform a given task. This allows the user to interact with a device according to their ability or preference. Users need options to choose methods that best fit their pace and experience. Systems that are flexible are more useful for a wide range of users.

5. Don't reinvent the wheel

It's important to innovate but not at the expense of learnability and accessibility. Users have developed old habits from other interfaces that will transfer to well-designed systems. A design that uses ready-made standards will be a more intuitive, useful product.

Principles to support usability

Learnability

the ease with which new users can begin effective interaction and achieve maximal performance

Flexibility

the multiplicity of ways the user and system exchange information

Robustness

the level of support provided the user in determining successful achievement and assessment of goal-directed behaviour



Principles of learnability

Predictability – users don't like surprises (exception games and then only a few)

- determining effect of future actions based on past interaction history
- operation visibility

Synthesizability – requires user to have a mental model

- assessing the effect of past actions
- immediate vs. eventual honesty – changing wysisyg doc vrs updating web pages

Familiarity

- how prior knowledge applies to new system
- guessability; affordance

Generalizability

- extending specific interaction knowledge to new situations

Consistency

- likeness in input/output behaviour arising from similar situations or task objectives



Principles of flexibility

Dialogue initiative – system and users in a conversation

- freedom from system imposed constraints on input dialogue
- system vs. user pre-emptiveness
- understanding of main use-cases

Multithreading

- ability of system to support user interaction for more than one task at a time
- concurrent vs. interleaving
- multimodality – button click / alt + / menu item

Task migratability

- passing responsibility for task execution between user and system
- ultimate user control

Substitutivity

- allowing equivalent values of input and output to be substituted for each other
- representation multiplicity (graph/values)
- equal opportunity (define line by drawing or specifying length/position)

Customizability

- modifiability of the user interface by user (adaptability) or system (adaptivity)



Principles of robustness

Observability

- ability of user to evaluate the internal state of the system from its perceivable representation
- browsability; defaults; reachability; persistence; operation visibility

Recoverability

- ability of user to take corrective action once an error has been recognized
- reachability; forward/backward recovery; commensurate effort

Responsiveness

- how the user perceives the rate of communication with the system
- Stability

Task conformance

- degree to which system services support all of the user's tasks
- task completeness; task adequacy



Standards

- set by national or international bodies to ensure compliance by a large community of designers standards require sound underlying theory and slowly changing technology
 - many large organisations have their own standards
- hardware standards more common than software high authority and low level of detail
- ISO 9241 defines usability as effectiveness, efficiency and satisfaction with which users accomplish tasks
- There are also some ISO standards for usability reporting



Nielsen's 10 Usability Heuristics

1. *Visibility of system status*
2. *Match between system and the real world*
3. *User control and freedom*
4. *Consistency and standards*
5. *Error prevention*
6. *Recognition rather than recall*
7. *Flexibility and efficiency of use*
8. *Aesthetic and minimalist design*
9. *Help users recognize, diagnose, and recover from errors*
10. *Help and documentation*



Shneiderman's 8 Golden Rules

- 1. Strive for consistency*
- 2. Enable frequent users to use shortcuts*
- 3. Offer informative feedback*
- 4. Design dialogs to yield closure*
- 5. Offer error prevention and simple error handling*
- 6. Permit easy reversal of actions*
- 7. Support internal locus of control*
- 8. Reduce short-term memory load*





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Implementation support





Implementation support by various Tools and OS

- programming tools
 - levels of services for programmers
- windowing systems
 - core support for separate and simultaneous user-system activity
- programming the application and control of dialogue
- interaction toolkits
 - bring programming closer to level of user perception
- user interface management systems
 - controls relationship between presentation and functionality



How does HCI affect of the programmer?

Advances in coding have elevated programming
hardware specific
→ interaction-technique specific

Layers of development tools

- windowing systems
- interaction toolkits
- user interface management systems

Elements of windowing systems

Device independence

programming the abstract terminal device
drivers

image models for output and (partially) input

- pixels
- PostScript (MacOS X, NextStep)
- Graphical Kernel System (GKS)
- Programmers' Hierarchical Interface to Graphics (PHIGS)

Resource sharing

achieving simultaneity of
user tasks

window system supports
independent processes

isolation of individual
applications



Architectures of windowing systems

three possible software architectures

- all assume device driver is separate
- differ in how multiple application management is implemented

1. each application manages all processes

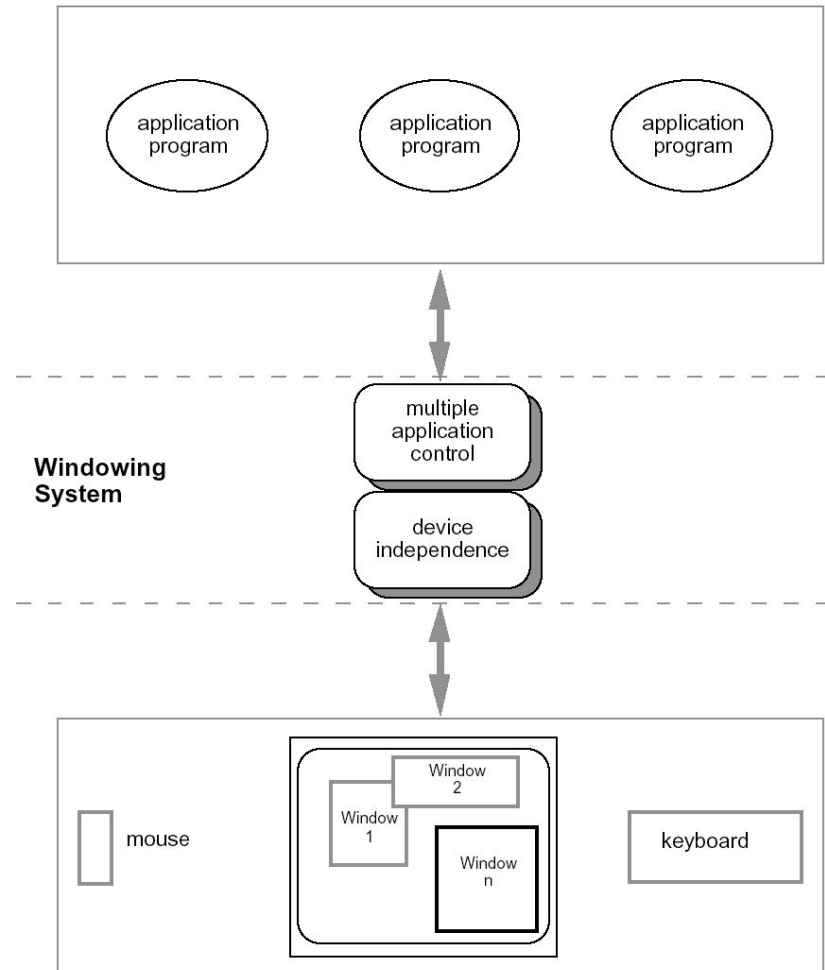
- everyone worries about synchronization
- reduces portability of applications

2. management role within kernel of operating system

- applications tied to operating system

3. management role as separate application
maximum portability

Roles of a windowing system

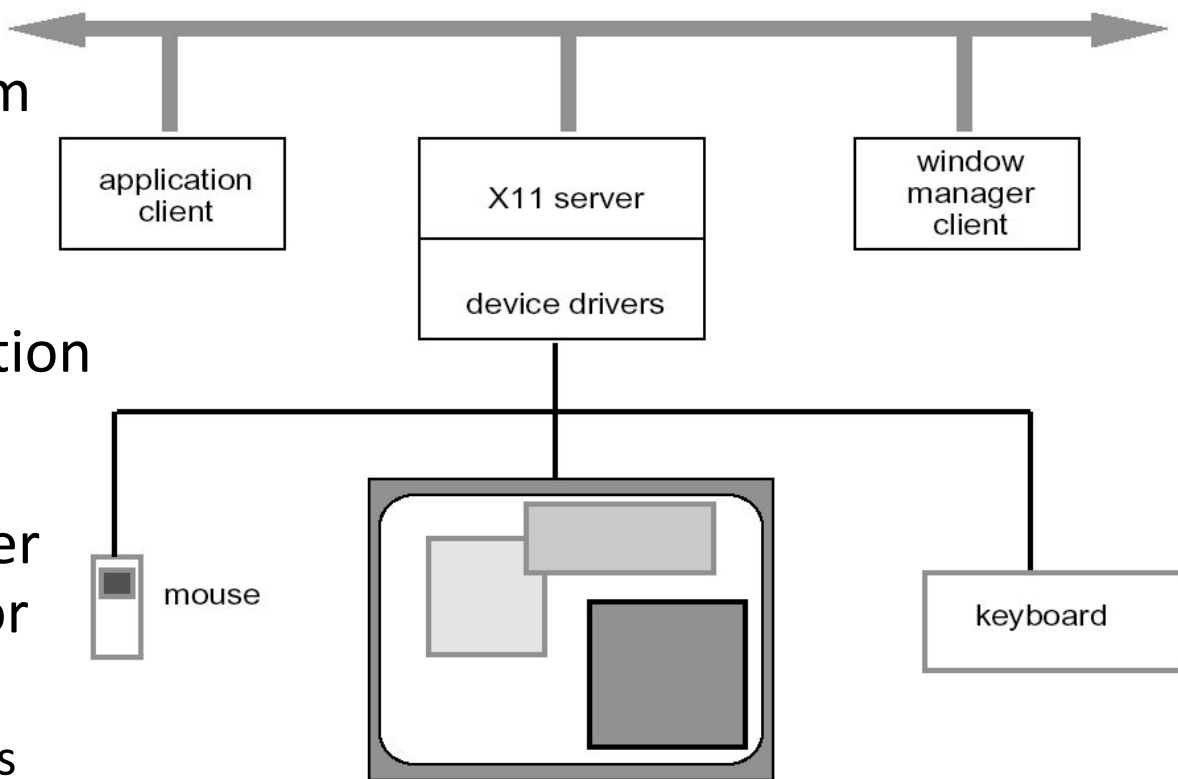


X Windows architecture



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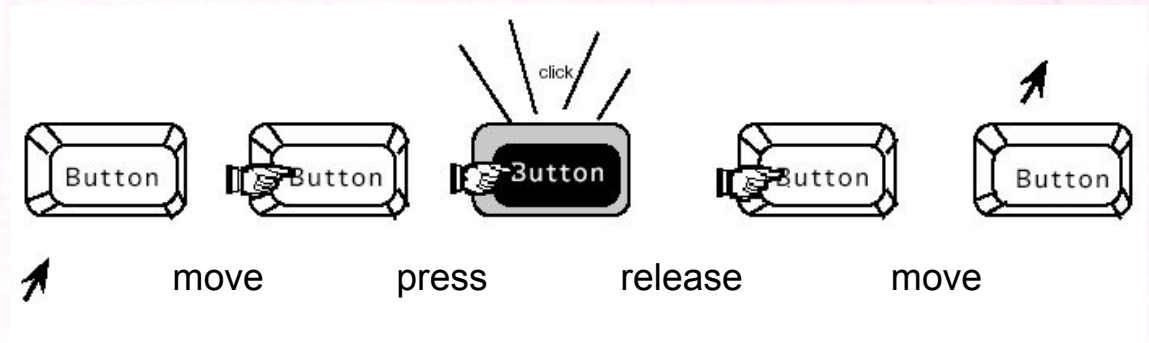
- pixel imaging model with some pointing mechanism
- X protocol defines server-client communication
- separate window manager client enforces policies for input/output:
 - how to change input focus
 - tiled vs. overlapping windows
 - inter-client data transfer



Using toolkits

Interaction objects

- input and output intrinsically linked



Toolkits provide this level of abstraction

- programming with interaction objects (or techniques, widgets, gadgets)
- promote consistency and generalizability
- through similar look and feel
- amenable to object-oriented programming



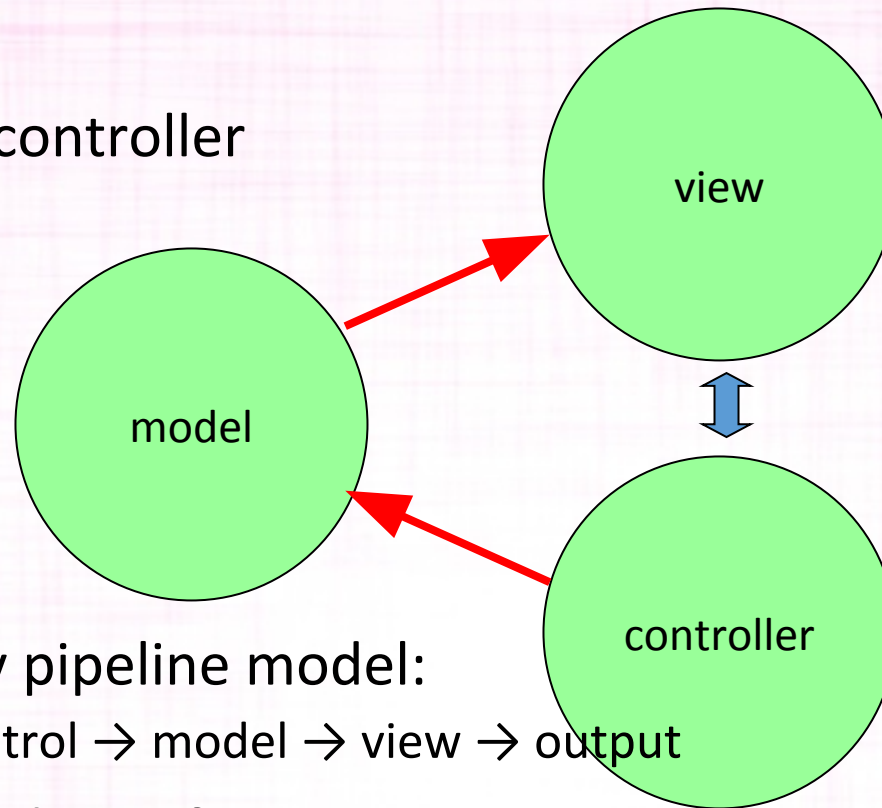
User Interface Management Systems (UIMS)

- UIMS add another level above toolkits
 - toolkits too difficult for non-programmers
- concerns of UIMS
 - conceptual architecture
 - implementation techniques
 - support infrastructure
- non-UIMS terms:
 - UI development system (UIDS)
 - UI development environment (UIDE)
 - e.g. Visual Basic
- *separation* between application semantics and presentation
- improves:
 - portability – runs on different systems
 - reusability – components reused cutting costs
 - multiple interfaces – accessing same functionality
 - customizability – by designer and user



MVC

model - view - controller

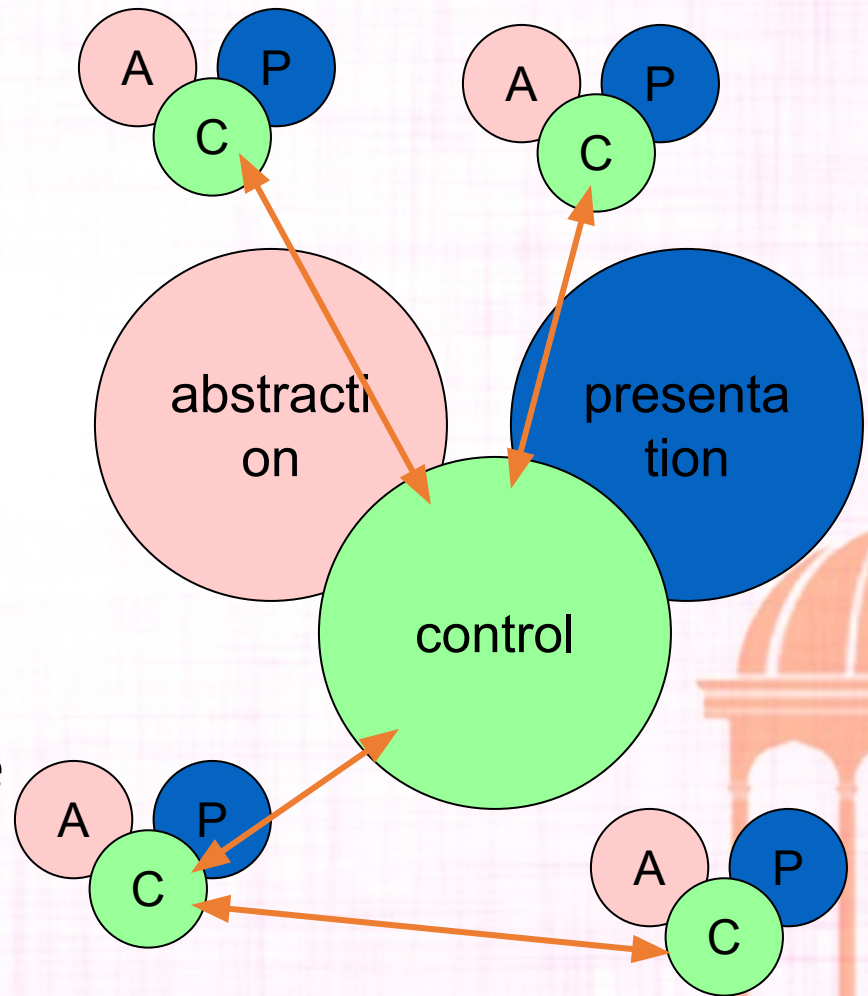


- MVC is largely pipeline model:
input → control → model → view → output
- but in graphical interface
 - input only has meaning in relation to output
- e.g. mouse click
 - need to know *what* was clicked
 - controller has to decide what to do with click
 - but view knows what is shown where!
- in practice controller ‘talks’ to view
 - separation not complete



PAC model

- PAC model closer to Seeheim
 - abstraction – logical state of component
 - presentation – manages input and output
 - control – mediates between them
- manages hierarchy and multiple views
 - control part of PAC objects communicate
- PAC cleaner in many ways ...
but MVC used more in practice
(e.g. Java Swing)





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Evaluation techniques



Evaluation Techniques

- Evaluation

- tests usability and functionality of system
- occurs in laboratory, field and/or in collaboration with users
- evaluates both design and implementation
- should be considered at all stages in the design life cycle

Goals of Evaluation

- assess extent of system functionality
- assess effect of interface on user
- identify specific problems

Evaluating Designs

- Cognitive Walkthrough
- Heuristic Evaluation
- Review-based evaluation



1. Cognitive Walkthrough

Proposed by Polson *et al.*



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- evaluates design on how well it supports user in learning task
- usually performed by expert in cognitive psychology
- expert 'walks through' design to identify potential problems using psychological principles
- forms used to guide analysis
- For each task walkthrough considers
 - what impact will interaction have on user?
 - what cognitive processes are required?
 - what learning problems may occur?
- Analysis focuses on goals and knowledge: does the design lead the user to generate the correct goals?



2. Heuristic Evaluation

- Proposed by Nielsen and Molich.
- usability criteria (heuristics) are identified
- design examined by experts to see if these are violated
- Example heuristics
 - system behaviour is predictable
 - system behaviour is consistent
 - feedback is provided
- Heuristic evaluation 'debugs' design.



3. Review-based evaluation

- Results from the literature used to support or refute parts of design.
- Care needed to ensure results are transferable to new design.
- Model-based evaluation
- Cognitive models used to filter design options
e.g. GOMS prediction of user performance. GOMS=
- Design rationale can also provide useful evaluation information



Evaluating through user Participation

1. Laboratory studies

- Advantages:
 - specialist equipment available
 - uninterrupted environment
- Disadvantages:
 - lack of context
 - difficult to observe several users cooperating
- Appropriate
 - if system location is dangerous or impractical for constrained single user systems to allow controlled manipulation of use



2. Field Studies

- **Advantages:**
 - natural environment
 - context retained (though observation may alter it)
 - longitudinal studies possible
- **Disadvantages:**
 - distractions
 - noise
- **Appropriate**
 - where context is crucial for longitudinal studies



3. Experimental evaluation

- controlled evaluation of specific aspects of interactive behaviour
- evaluator chooses hypothesis to be tested
- a number of experimental conditions are considered which differ only in the value of some controlled variable.
- changes in behavioural measure are attributed to different conditions





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Universal Design



Universal design principles

- equitable use
- flexibility in use
- simple and intuitive to use
- perceptible information
- tolerance for error
- low physical effort
- size and space for approach and use



Multi-Sensory Systems



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- More than one sensory channel in interaction
 - e.g. sounds, text, hypertext, animation, video, gestures, vision
- Used in a range of applications:
 - particularly good for users with special needs, and virtual reality
- Will cover
 - general terminology
 - speech
 - non-speech sounds
 - handwriting
- considering applications as well as principles



- Multi-modal systems

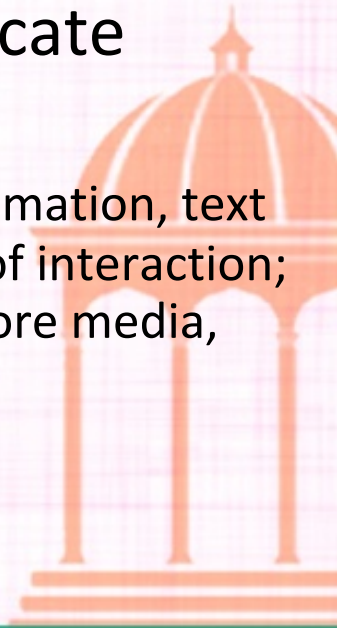
- use more than one sense (or mode) of interaction

e.g. visual and aural senses: a text processor may speak the words as well as echoing them to the screen

- Multi-media systems

- use a number of different media to communicate information

e.g. a computer-based teaching system: may use video, animation, text and still images: different media all using the visual mode of interaction; may also use sounds, both speech and non-speech: two more media, now using a different mode





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User Support



- **Issues**
 - different types of support at different times
 - implementation and presentation both important
 - all need careful design
- **Types of user support**
 - quick reference, task specific help, full explanation, tutorial
- **Provided by help and documentation**
 - help - problem-oriented and specific
 - documentation - system-oriented and general
 - same design principles apply to both

Requirements



- **Availability**
 - continuous access concurrent to main application
- **Accuracy and completeness**
 - help matches and covers actual system behaviour
- **Consistency**
 - between different parts of the help system and paper documentation
- **Robustness**
 - correct error handling and unpredictable behaviour
- **Flexibility**
 - allows user to interact in a way appropriate to experience and task
- **Unobtrusiveness**
 - does not prevent the user continuing with work



Approaches to user support

- Command assistance
 - User requests help on particular command
e.g., UNIX man, DOS help
 - Good for quick reference
 - Assumes user know what to look for
- Command prompts
 - Provide information about correct usage when an error occurs
 - Good for simple syntactic errors
 - Also assumes knowledge of the command
- Context sensitive help
 - help request interpreted according to context in which it occurs. e.g. tooltips
- On-line tutorials
 - user works through basics of application in a test environment.
 - can be useful but are often in flexible.
- On-line documentation
 - paper documentation is made available on computer.
 - continually available in common medium
 - can be difficult to browse
 - hypertext used to support browsing.



Wizards and Assistants

- wizards

- task specific tool leads the user through task, step by step, using user's answers to specific questions
- example: resumé
- useful for safe completion of complex or infrequent tasks
- constrained task execution so limited flexibility
- must allow user to go back

- assistants

- monitor user behaviour and offer contextual advice
- can be irritating e.g. MS paperclip
- must be under user control e.g. XP smart tags
- Virtual assistants
- Chat bots



Knowledge representation

1. Domain and task modelling

Covers

- common errors and tasks
- current task

Usually involves analysis of command sequences.

Problems

- representing tasks
- interleaved tasks
- user intention

2. Advisory strategy

involves choosing the correct style of advice for a given situation.
e.g. reminder, tutorial, etc.

few intelligent help systems model advisory strategy, but choice of strategy is still important.



Techniques for knowledge representation

- rule based (e.g. logic, production rules)
 - knowledge presented as rules and facts
 - interpreted using inference mechanism
 - can be used in relatively large domains.
- frame based (e.g. semantic network)
 - knowledge stored in structures with slots to be filled
 - useful for a small domain.
- network based
 - knowledge represented as relationships between facts
 - can be used to link frames.
- example based
 - knowledge represented implicitly within decision structure
 - trained to classify rather than programmed with rules
 - requires little knowledge acquisition





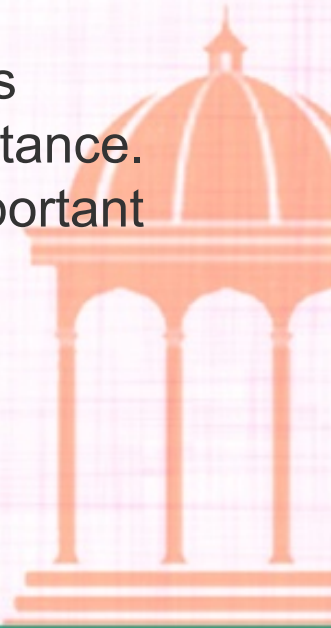
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Designing interfaces for all



Best Practices for Designing an Interface

- **Keep the interface simple.** The best interfaces are almost invisible to the user. They avoid unnecessary elements and are clear in the language they use on labels and in messaging.
- **Create consistency and use common UI elements.** By using common elements in your UI, users feel more comfortable and are able to get things done more quickly. It is also important to create patterns in language, layout and design throughout the site to help facilitate efficiency. Once a user learns how to do something, they should be able to transfer that skill to other parts of the site.
- **Be purposeful in page layout.** Consider the spatial relationships between items on the page and structure the page based on importance. Careful placement of items can help draw attention to the most important pieces of information and can aid scanning and readability.



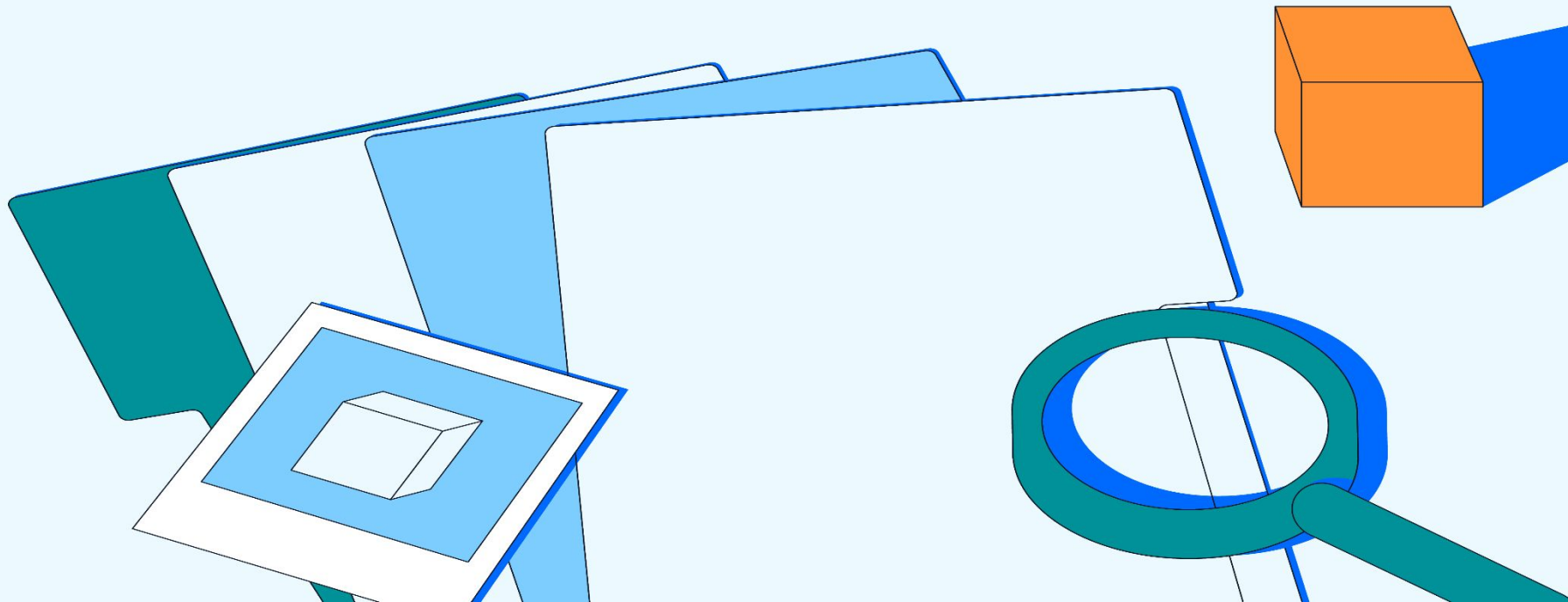
Best Practices for Designing an Interface

- **Strategically use color and texture.** You can direct attention toward or redirect attention away from items using color, light, contrast, and texture to your advantage.
- **Use typography to create hierarchy and clarity.** Carefully consider how you use typeface. Different sizes, fonts, and arrangement of the text to help increase scalability, legibility and readability.
- **Make sure that the system communicates what's happening.** Always inform your users of location, actions, changes in state, or errors. The use of various UI elements to communicate status and, if necessary, next steps can reduce frustration for your user.
- **Think about the defaults.** By carefully thinking about and anticipating the goals people bring to your site, you can create defaults that reduce the burden on the user. This becomes particularly important when it comes to form design where you might have an opportunity to have some fields pre-chosen or filled out.



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User research and techniques





User experience (UX) research methods are different ways of generating insights about your users, their behavior, motivations, and needs. You can use methods like user interviews, surveys, focus groups, card sorting, usability testing to identify user challenges and turn them into opportunities to improve the user experience.



The most common types of user research



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Qualitative vs. quantitative

Qualitative research focuses on capturing subjective insights into users' experiences. It aims to understand the underlying **reasons, motivations, and behaviors** of individuals. Quantitative research, on the other hand, involves **collecting and analyzing numerical data** to identify patterns, trends, and significance. It aims to quantify user behaviors, preferences, and attitudes, allowing for generalizations and statistical insights.

Attitudinal vs. behavioural

Attitudinal research is about understanding users' attitudes, perceptions, and beliefs. It delves into the 'why' behind user decisions and actions. It often involves surveys or interviews where users are asked about their **feelings, preferences, or perceptions towards a product or service**. It's subjective in nature, aiming to capture people's emotions and opinions.

Behavioral research is about what users do rather than what they say they do or would do. This kind of research is often based on observation methods like usability testing, eye-tracking, or heat maps to understand user behavior.

The most common types of user research

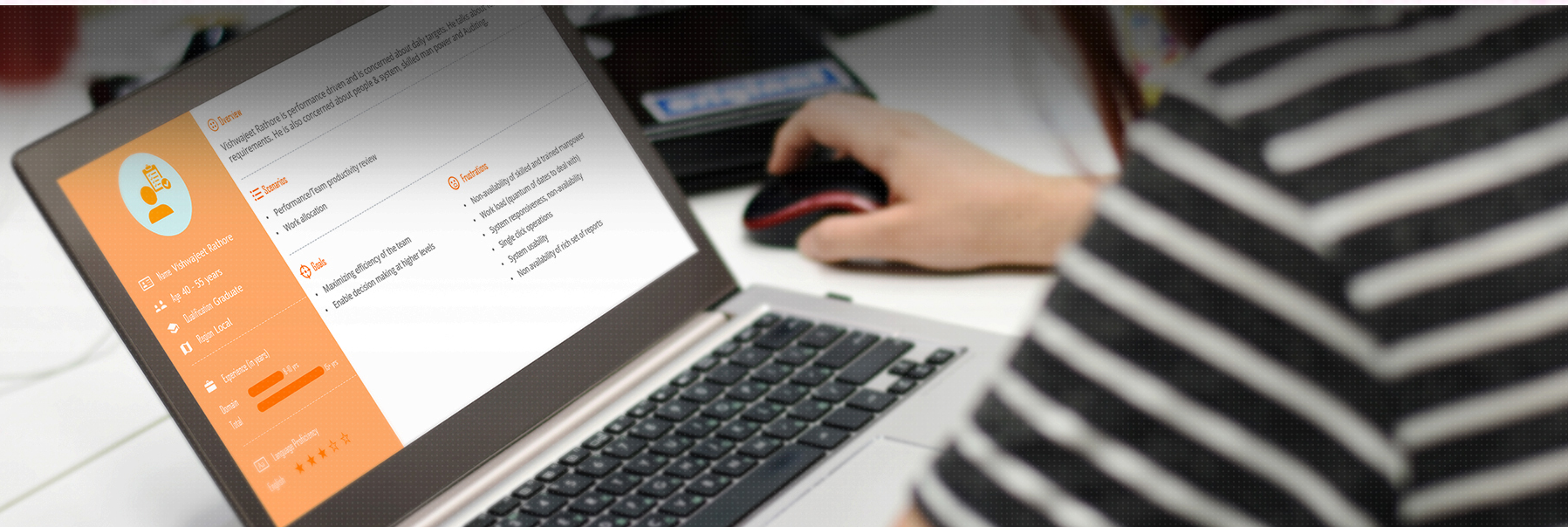
Generative vs. evaluative

Generative research is all about generating new ideas, concepts, and insights to fuel the design process. You might run brainstorming sessions with groups of users, card sorting, and co-design sessions to inspire creativity and guide the development of user-centered solutions.

On the other hand, **Evaluative research** focuses on assessing the usability, effectiveness, and overall quality of existing designs or prototypes. Once you've developed a prototype of your product, it's time to evaluate its strengths and weaknesses. You can compare different versions of a design or feature through A/B testing—ensuring your design meets user needs and expectations.

Research technique	Description	When to use it	Best for gathering
User interviews	One-on-one open-ended and guided discussions	Start and end of your project	Qualitative Generative
Field studies	Observe people in their natural environment	All stages	Qualitative Behavioral
Focus group	Group discussions facilitated by a moderator	Start and end of your project	Qualitative Generative
Diary studies	Users keep a diary to track interactions and experience with a product	Start of your project	Qualitative Evaluative
Surveys	Asking people open or closed questions	All stages	Qualitative Quantitative Attitudinal Generative Evaluative
Card sorting	Users sort information and ideas into groups that makes sense to them	Start of your project	Qualitative Generative Attitudinal
Tree testing	Assess the findability and organization of information as users navigate a stripped-down IA	Start of your design or redesign process	Quantitative Behavioral Evaluative
Usability testing	Users perform a set of tasks in a controlled setting	All stages	Qualitative Behavioral Evaluative
Five second testing	Collect immediate impressions within a short timeframe	During initial ideation and throughout design	Attitudinal Evaluative
A/B testing	Compare two versions of a solution	All stages	Quantitative Evaluative

Understanding Persona





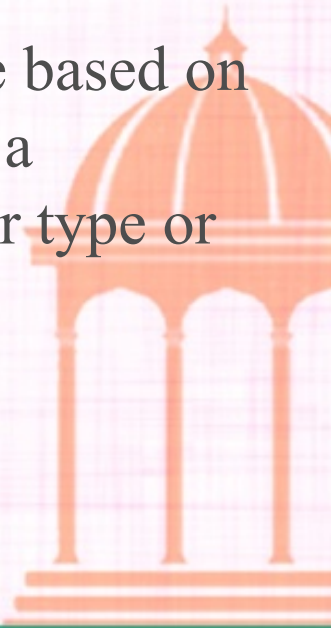
Personas are character profiles, which are created based on user research for products and services. The user research helps in classifying different user types who have similar behavior.

These user types are then described in detail with their attributes. These representative descriptions are personas. Creating personas helps understand ones users with their needs, behavior, and expectations. Personas enable researchers to create a picture of whom they are designing for. This makes the design process much easier.



More about Personas

- Essentially, a persona is a model, a summary profile of user types. It is outlined as a detailed description of a specific person; however, this person is fictional.
- Personas help researchers focus on specific characters rather than diffuse efforts to design for a large number of users.
- Personas have become increasingly popular in designing solutions that are user centered than designer.
- Even though personas do not describe real people, they are based on real data from real people, collected and then collated into a summarized character that has generic attributes of the user type or group, one is trying to describe.





Advantages of Personas

1. Human centered

The process of creating and the final personas that guide the design process take into account the needs and goals of the users.

2. Simpler design process

After the Personas are created, they act as the guiding principles for the rest of the design process, making the job of the designer easier as the designer becomes aware of whom they are designing for.

3. Higher acceptance by users

As the design is in accordance with who the users are and their complete descriptions taking into account their needs and goals, the final solution is as close to what the user needs and desires which gets a higher acceptance as a result.



Disadvantages of Personas

1. Generic description & missed details

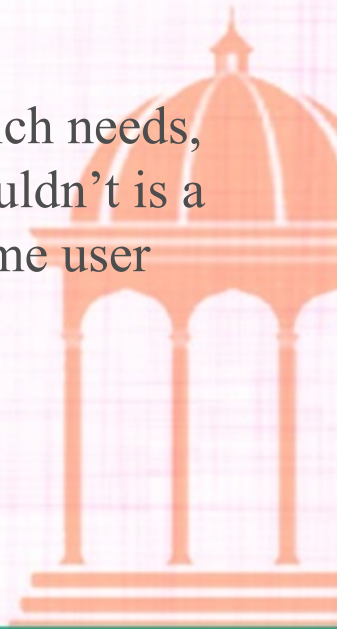
Personas are after all, a general description of a cluster of users who are grouped together based on their similar behavior, needs or goals. Therefore, not all attributes of each individual user that is part of this cluster get covered. This may lead to an important aspect of individual users getting missed, which may be important to them.

2. Time consuming

Creating personas is a time consuming and elaborate process. Time is require to first recruit users, research users, cluster them and then create personas based on that research.

3. Complex findings

Every user may have different needs, behavior, goals and prioritizing which needs, behavior patterns and goals should be part of the personas and which shouldn't is a huge task in itself. Also, personas should incorporate exceptions or extreme user descriptions to cover all aspects of users one is designing for.





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Ergonomics



Ergonomics Definition

Ergonomics

Ergonomics (or human factors) is the scientific discipline concerned with the understanding of interactions among humans and other elements of a system, and the profession that applies theory, principles, data and methods to design in order to optimize human well-being and overall system performance.

The word ergonomics comes from the Greek word “ergon” which means work and “nomos” which means laws. It’s essentially the “laws of work” or “science of work”. **Good ergonomic design** removes incompatibilities between the work and the worker and creates the optimal work environment.



Ergonomics draws on many disciplines to optimize the interaction between the work environment and the worker.

Disciplines

- Anthropometry
- Biomechanics
- Mechanical engineering
- Industrial engineering
- Industrial design
- Information design
- Kinesiology
- Physiology
- Psychology



Ergonomics Domains of Specialization

According to the International Ergonomics Association, there are three broad **domains of ergonomics**: **physical, cognitive, and organizational.**

Physical Ergonomics

Physical ergonomics is concerned with human anatomical, anthropometric, physiological and biomechanical characteristics as they relate to physical activity.

This is the ergonomics domain we are most concerned with in the workplace, and most of the content on this site is very much focused on **workplace ergonomics.**





Workplace Ergonomics

The science of fitting workplace conditions and job demands to the capabilities of the working population. Ergonomics is an approach or solution to deal with a number of problems—among them are work-related musculoskeletal disorders.

At its core, workplace ergonomics is really about building a better workplace. When jobs are designed to match the capabilities of people, it results in better work being produced and a better experience for the person doing it.

Through that lens, ergonomics creates value on several fronts. It's good for your people and good for your business.



Benefits of Ergonomics

1. Lower costs
2. Higher productivity
3. Better product quality
4. Improved employee engagement
5. Better safety culture

The ergonomics improvement process systematically identifies ergonomic hazards and puts in place engineering and administrative control measures to quantifiably reduce risk factors.



Cognitive Ergonomics

Cognitive ergonomics is concerned with mental processes, such as perception, memory, reasoning, and motor response, as they affect interactions among humans and other elements of a system.

Relevant topics

- mental workload
- decision-making
- skilled performance
- human-computer interaction
- human reliability
- work stress
- training as these may relate to human-system design





Organizational Ergonomics

Organizational ergonomics is concerned with the optimization of sociotechnical systems, including their organizational structures, policies, and processes.

Relevant topics

- communication
- crew resource management
- work design
- design of working times
- teamwork
- participatory design
- community ergonomics
- cooperative work
- new work paradigms
- virtual organizations
- telework
- quality management





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Exercise - Creating personae for different applications in everyday use.

Activity : Create Your Own Personae For Daily Routine





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Thank You

